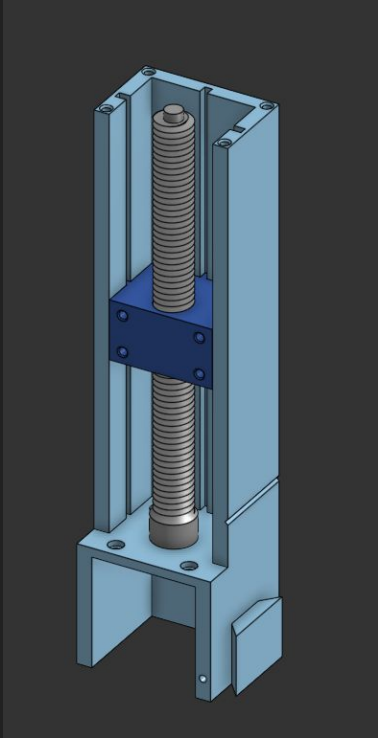


HexBot V2 Assembly Guide

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Z-Axis



Parts Needed:

1x lead screw (Printed at 96% scale on x and y axis, print test piece to find optimal sale)

1 Lead Screw housing

1x Lead Screw bearing block

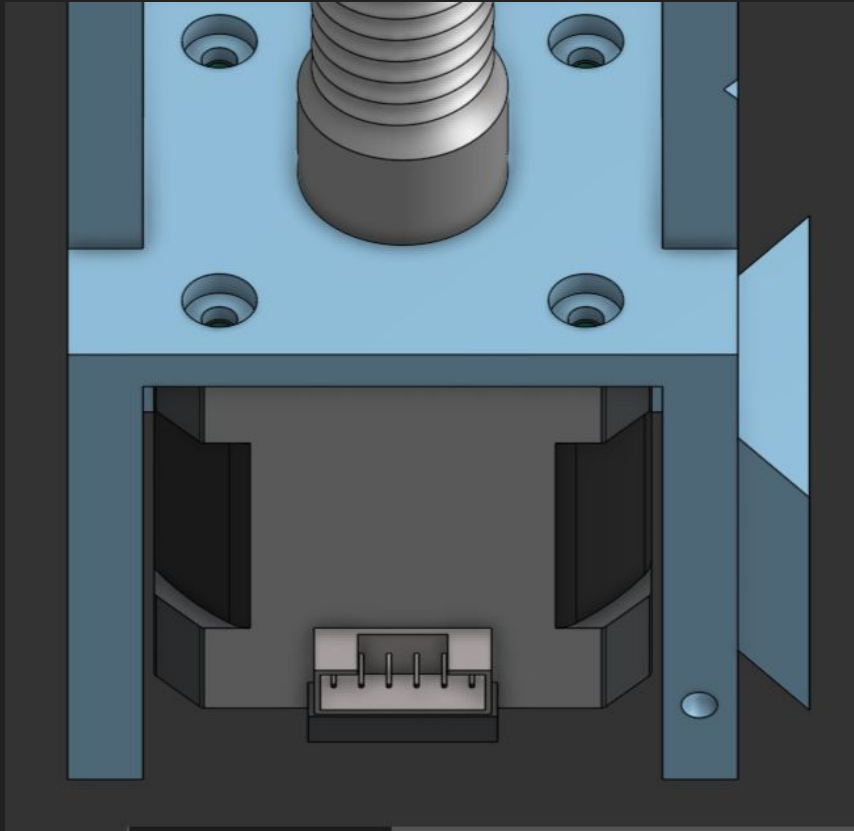
1x Lead screw cap & extruder mount

1x Stepper Motor (Nema 17)

9x M3 heat set inserts

8x M3 screws

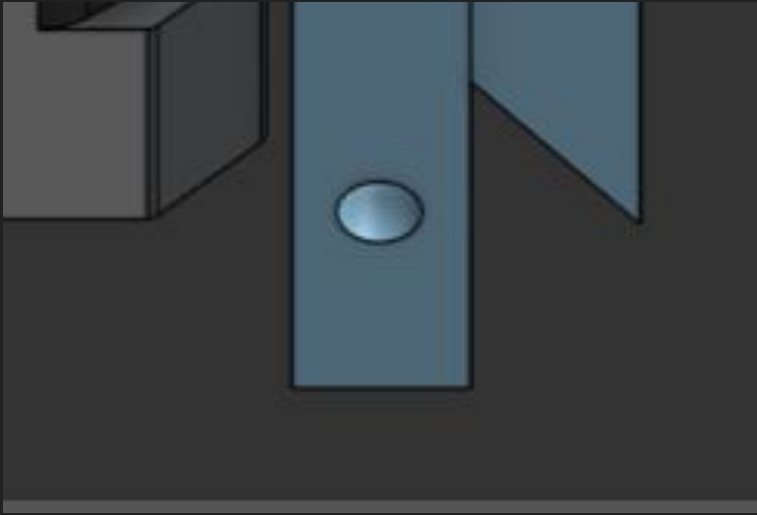
Step 1:



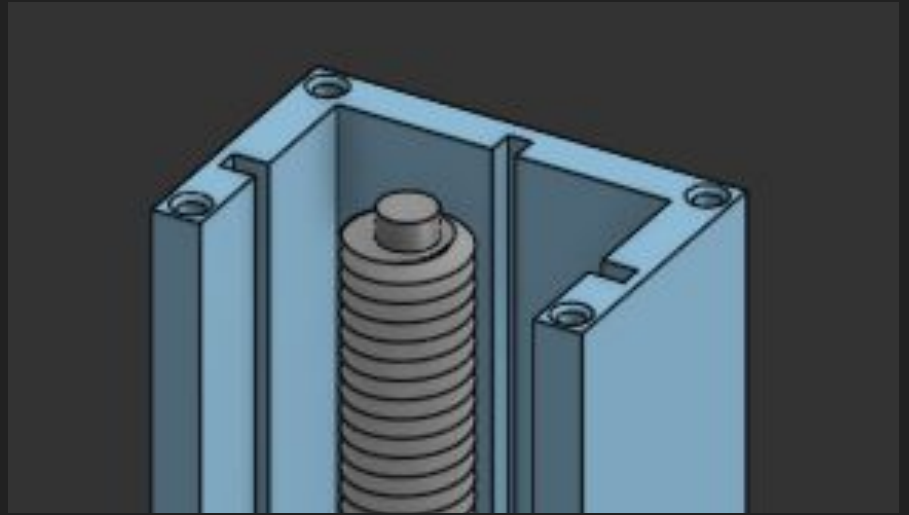
Take the Nema 17 and screw it in.

Make sure the connector/wires are facing out.

Step 2:

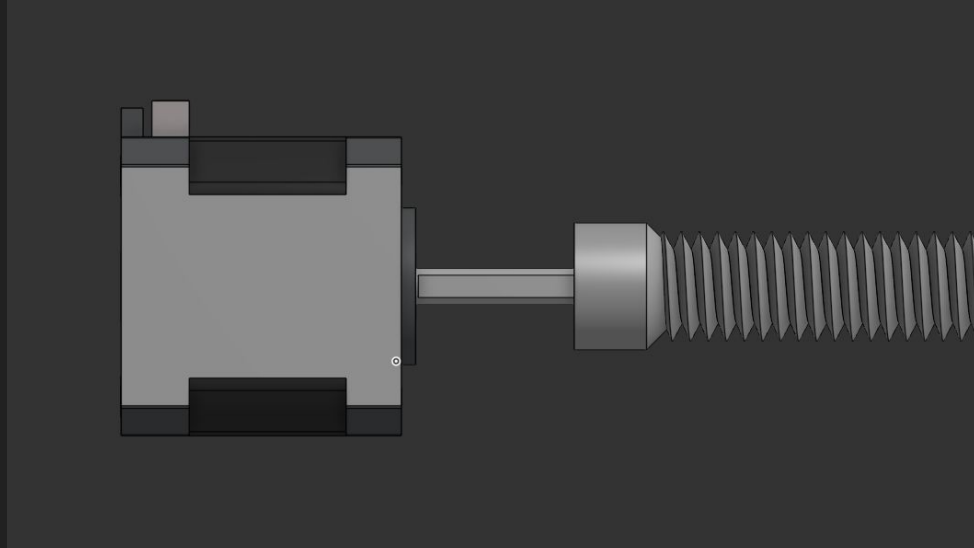


Put one heated insert here



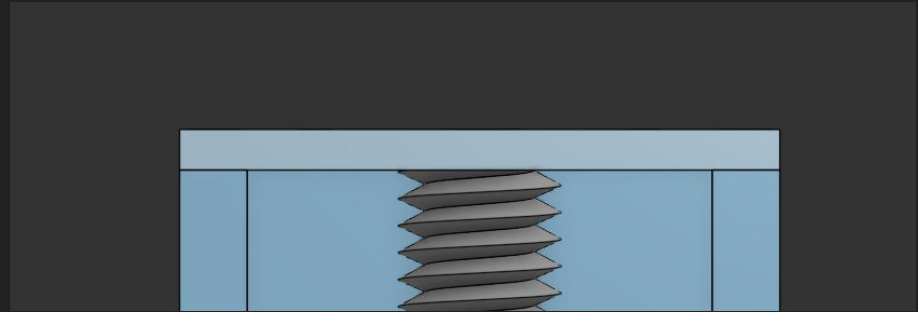
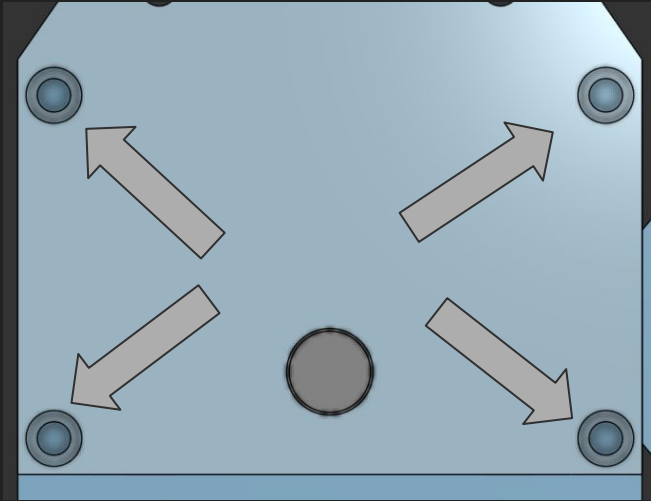
Put 4 heated inserts here

Step 3:



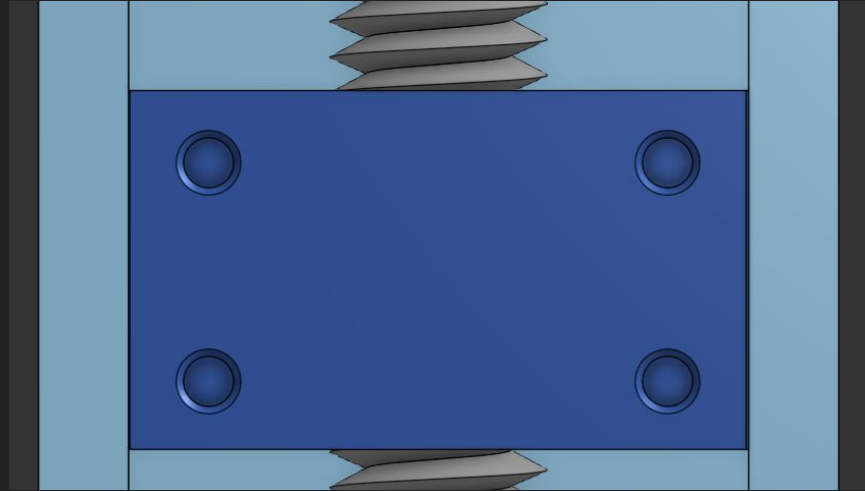
Slide Lead screw onto the
Nema 17 shaft.

Step 4:



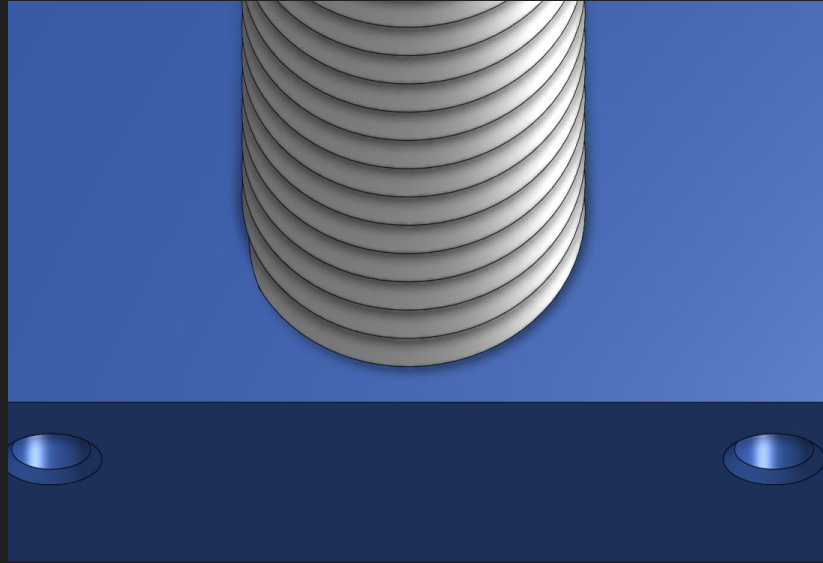
Put an M3 screw into the threaded insert with the lead screw top in the middle of the screw head and the insert.

Step 5:



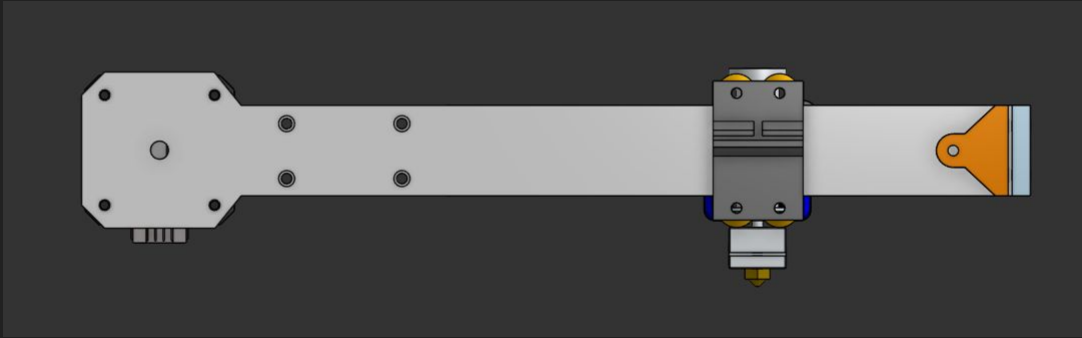
Put one M3 heat insert in each of the 4 holes

Step 6:



Screw the lead screw into the lead screw bearing block

X-Axis



Parts Needed:

1x Main Arm

1x Idler Holder

1x Idler tensioner

1x Belt Plate

1x Hotend Plate

2x rollers

1x bottom holder

1x Stepper Motor (Nema 17)

10x M3 heat set inserts

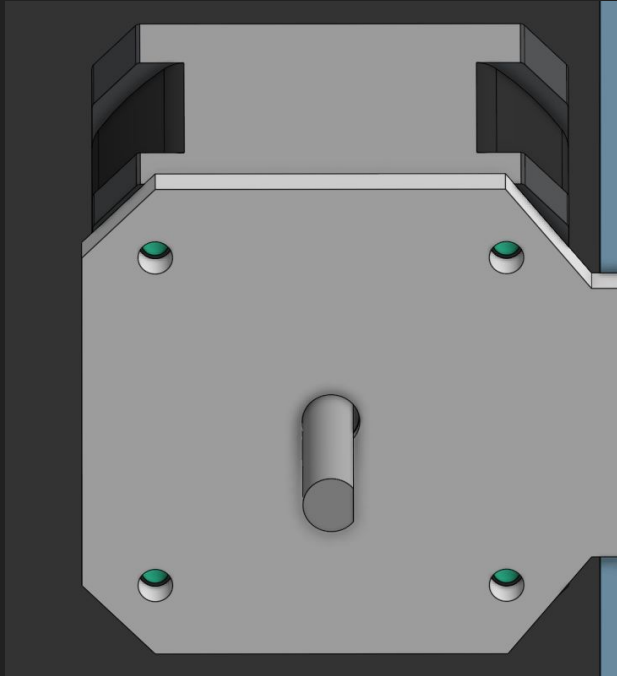
8x M3 screws

1x pulley & idler

1x 1 Meter GT2 timing belt

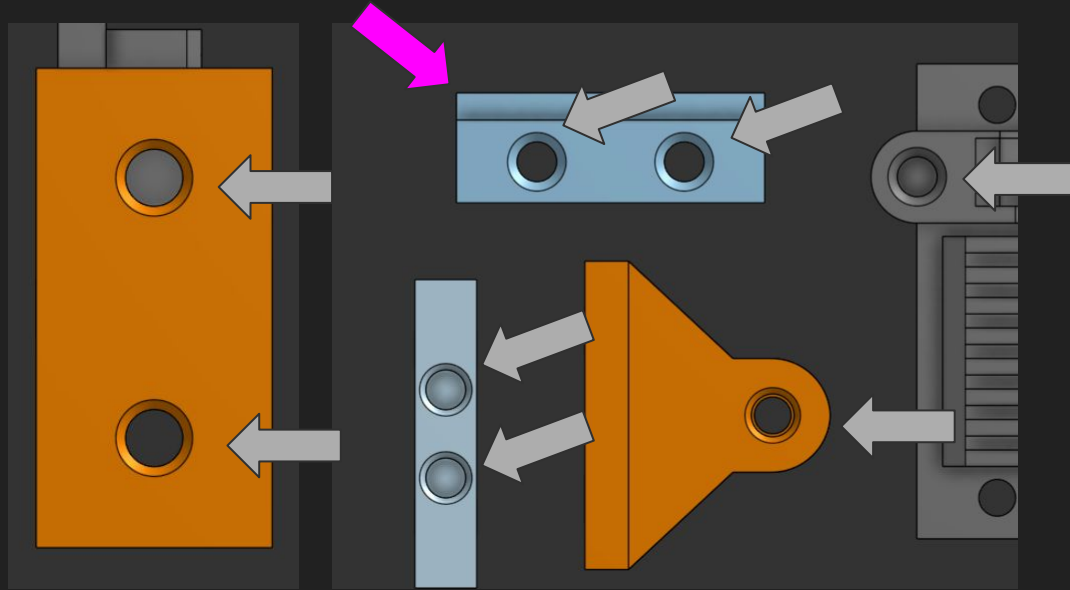
2x Pins for rollers

Step 1:



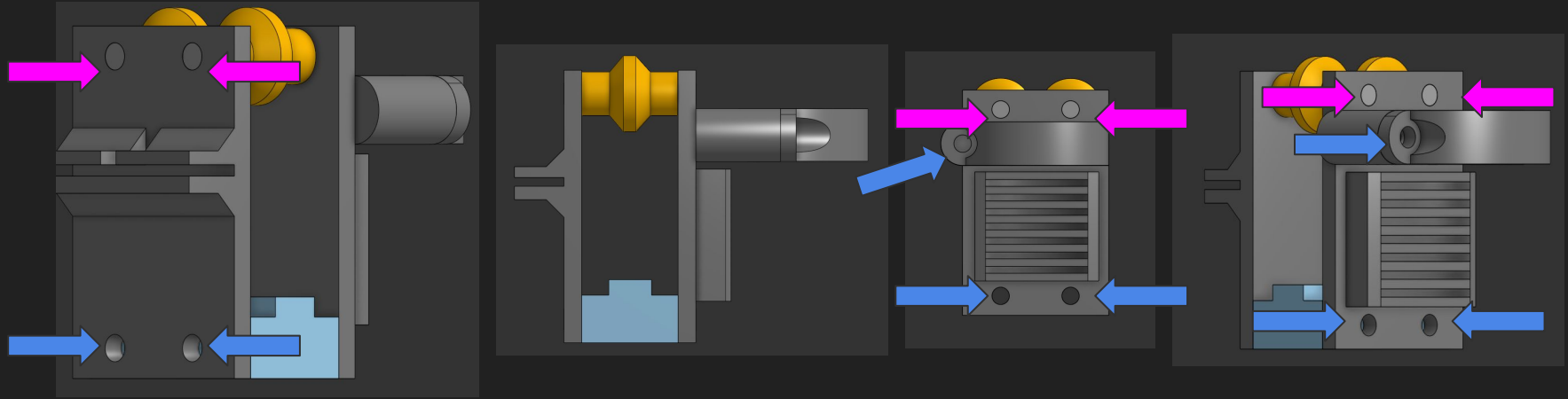
Screw the Nema 17 into the far left end of the arm

Step 2:



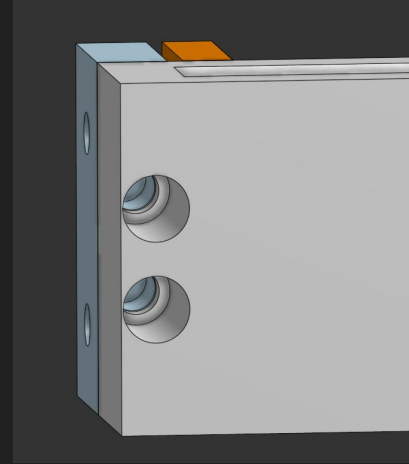
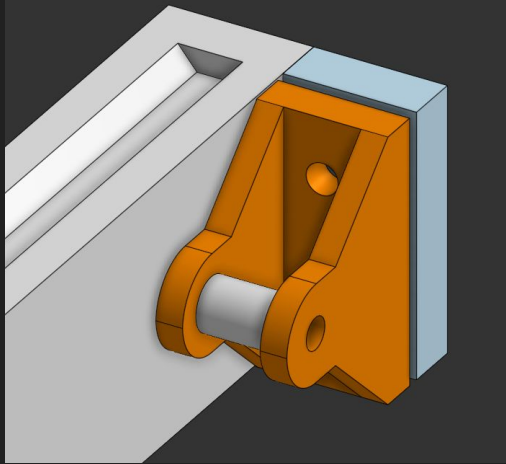
Put heated inserts in all the holes shown. The blue object labeled by the pink arrow requires both sides to have heat inserts. The orange part shown needs one on the side as shown in the right figure and two in the back as shown in the left.

Step 3:



Assemble the carriage so it looks like this. (Make sure to assemble it on the main white piece bc there is no way yo get it on after assembly without taking it apart.) The rods go from hole to hole marked by pink arrows. Screws are marked by blue arrows.

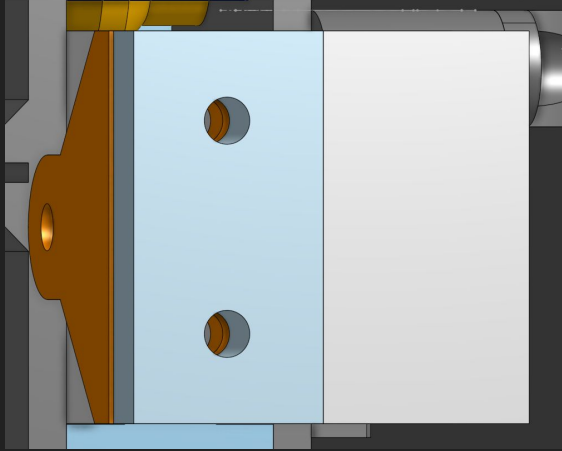
Step 4:



Take the idler holder (orange) and put the idler (White tube) in the middle of block (light blue) and put a hole. Put a M3 screw into the front side and screw until it reaches the heat insert in the back side.

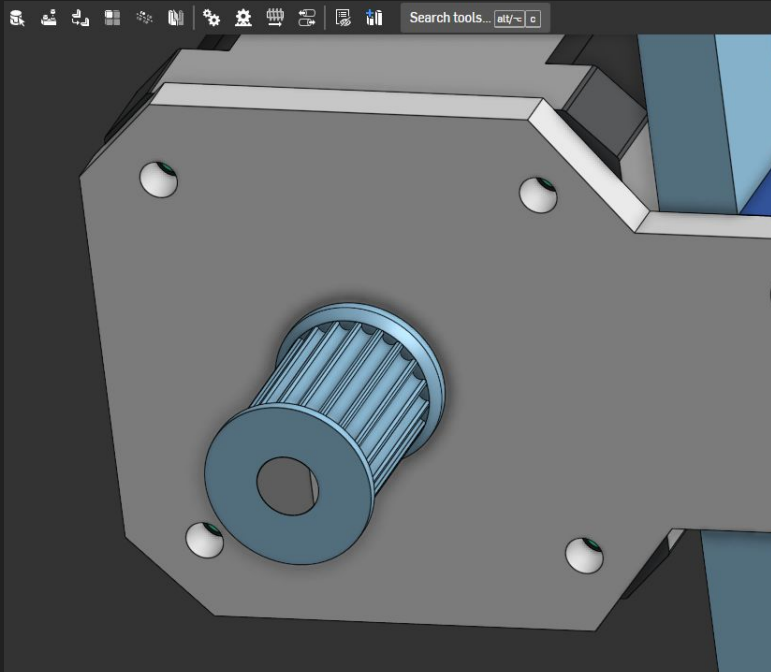
Take the idler tensioner screw through the rail (white)

Step 5:

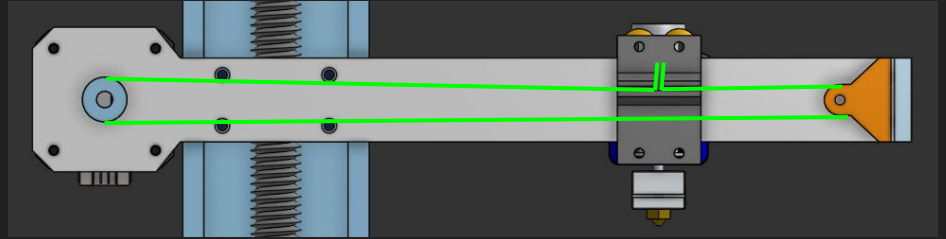


Put a screw through each hole on the light blue part and make sure it touches the threaded insert in the orange part. To tension the belt just turn the screws right.

Step 6:

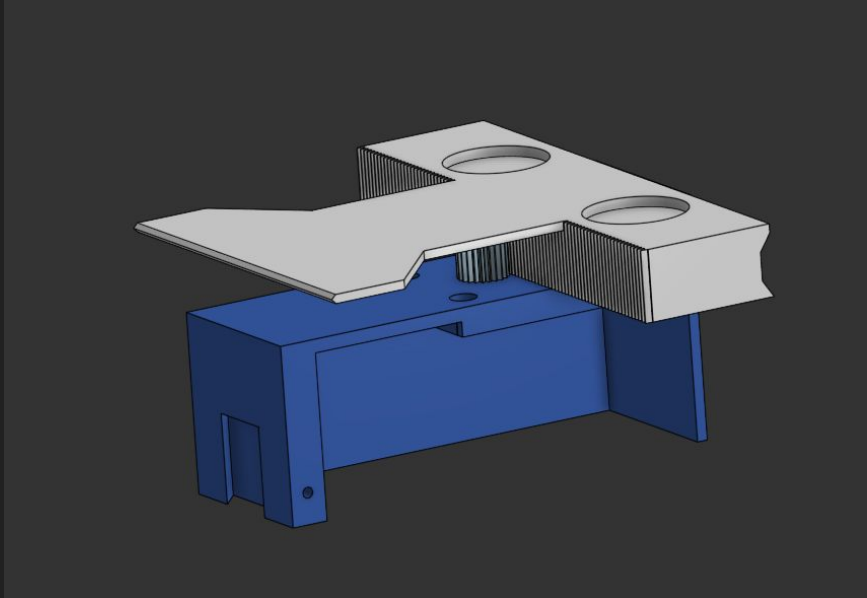


Put the Pulley on the shaft
of the NEMA 17



Run the GT2 timing belt as shown by the
green lines. Just cut off the excess belt
once you have it belted.

Y-Axis



Parts Needed:

1x Y-Motor mount

1x Bed Platform

1x Gear

1x Nema 17

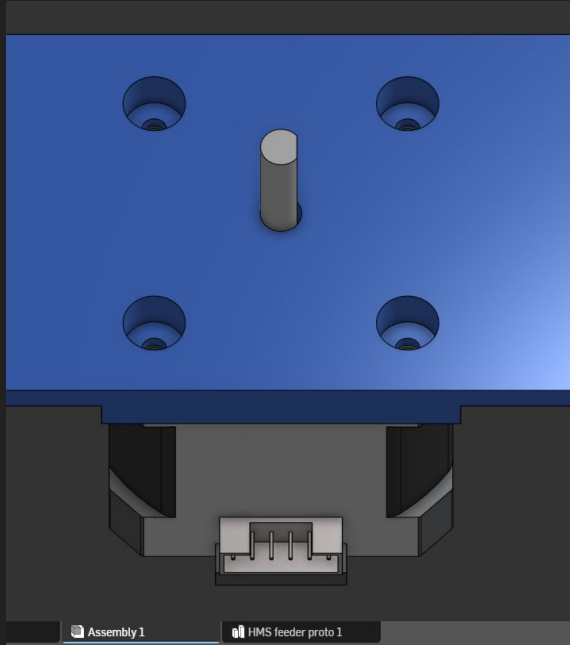
1x heat insert

1x 165x165 build plate (can be 3d printed as long as painters tape is applied to the top)

Optional :

2x Magnets less than 0.11 in tall

Step 1:

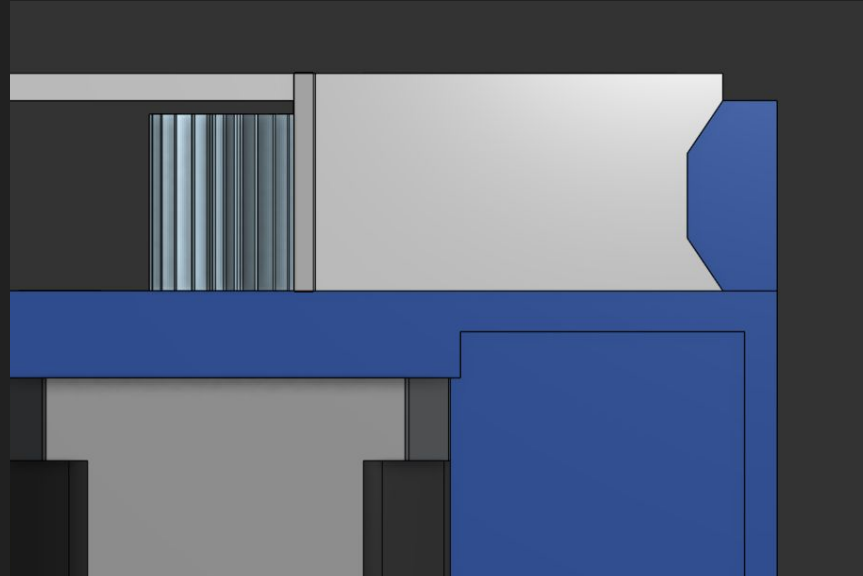


Screw the NEMA 17 in.
(Make sure the screws are
either flush or below the
top because that can lead
to interference.)

Step 2:

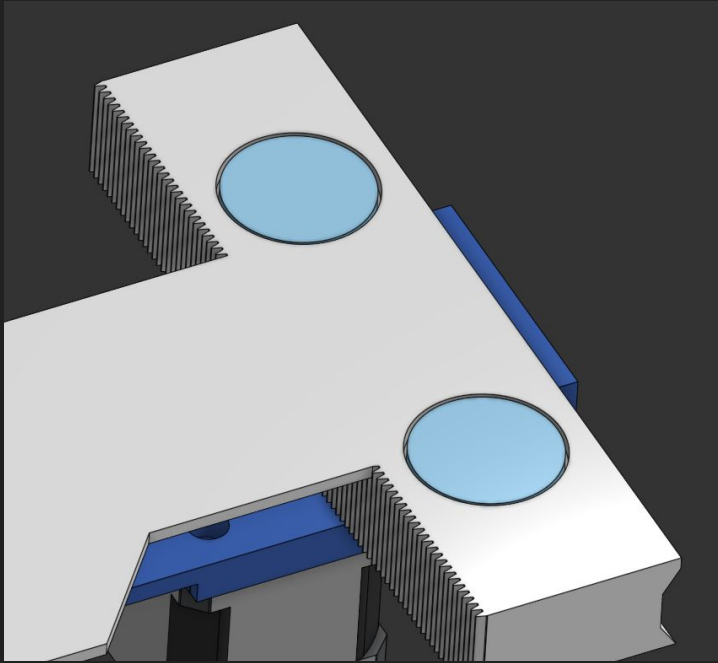


Put the gear on the Nema 17. (make sure flat side goes to flat side)



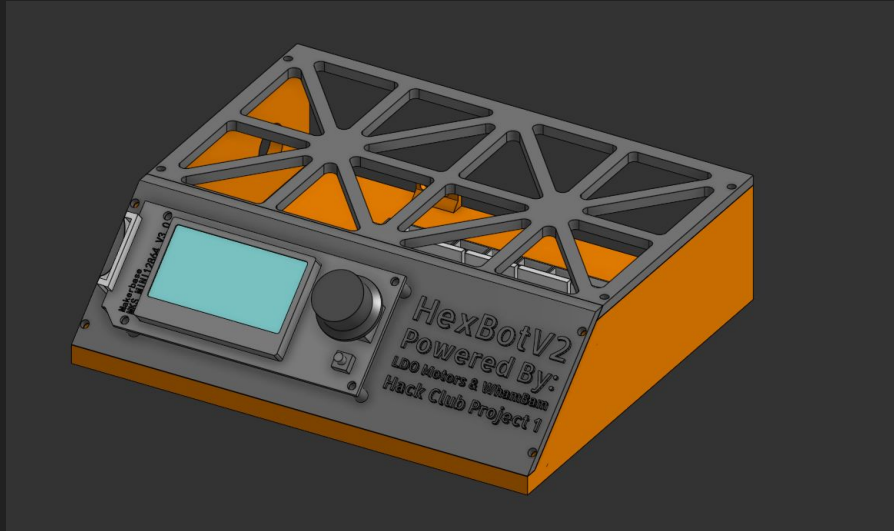
Slide the rack (white) on to the track.

Step 3: (Optional)



Put 2 magnets in the slots. No taller than 0.11 in (2.8mm) and no wider than 1.1 in (28mm) just use hot glue or sum to attach.

Base



Parts Needed:

1x front plate

1x top plate

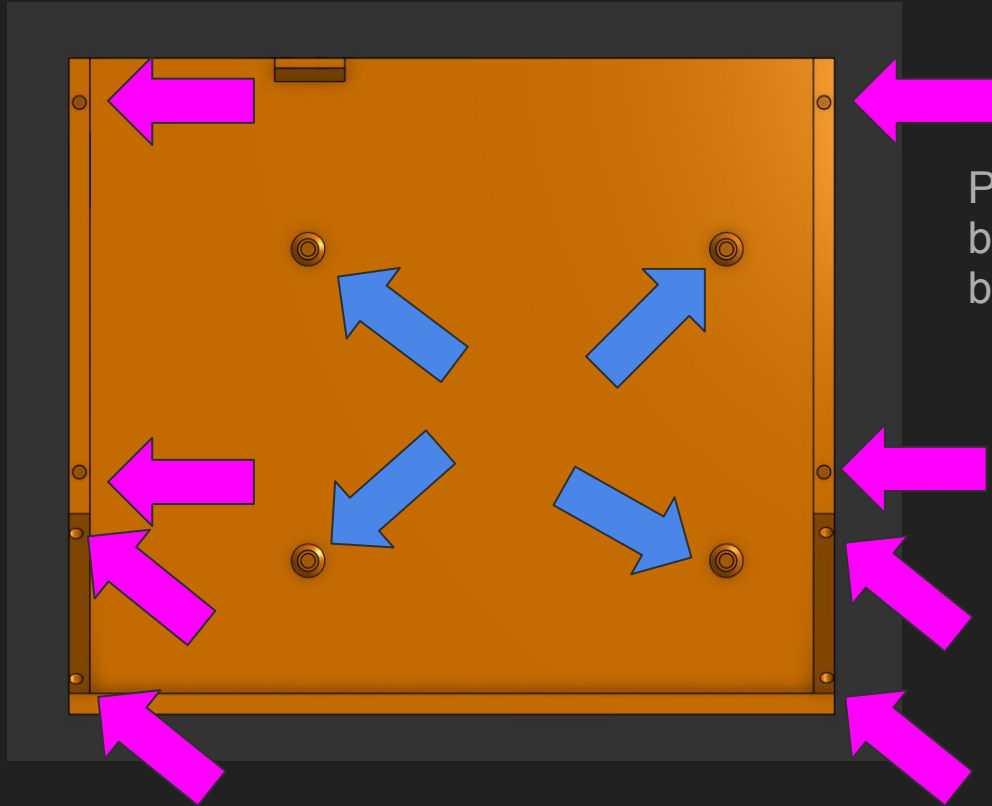
1x base tray

1x MKS Mini12864 V3 (screen)

1x MKS TinyBee V1.0 (Board)

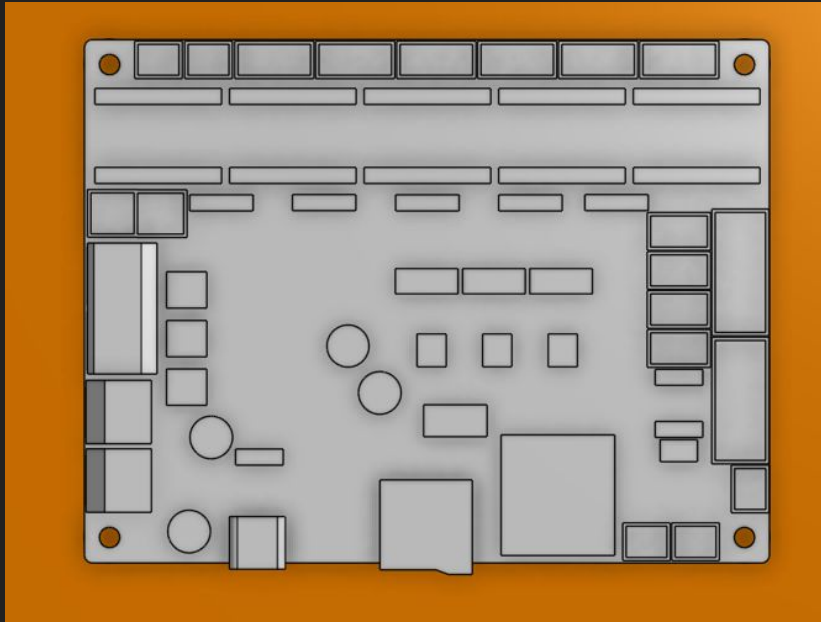
16x heat insert

Step 1:



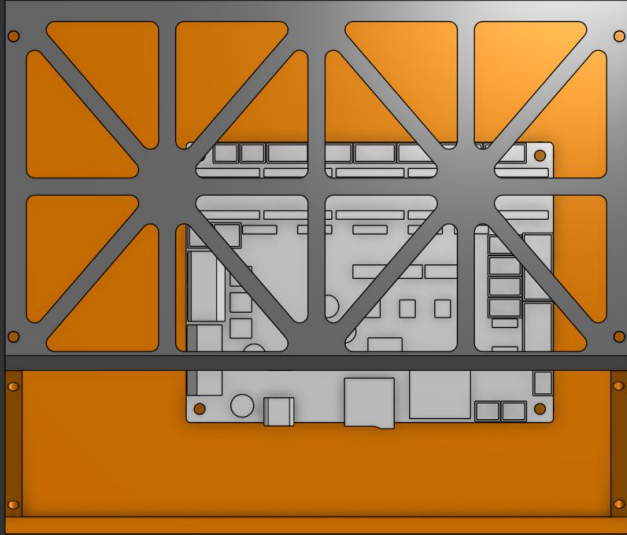
Put heated inserts into every hole marked by a pink arrow. The blue arrows need to be recessed in about 0.1 in (2.5mm).

Step 2:

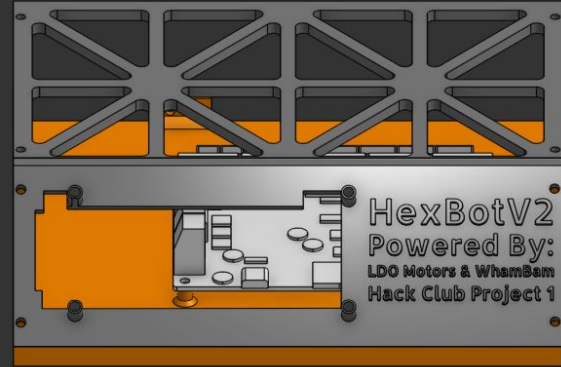


Put the MKS tinybee on. (Do Not overtighten bc I dont want u to break the board)

Step 3:

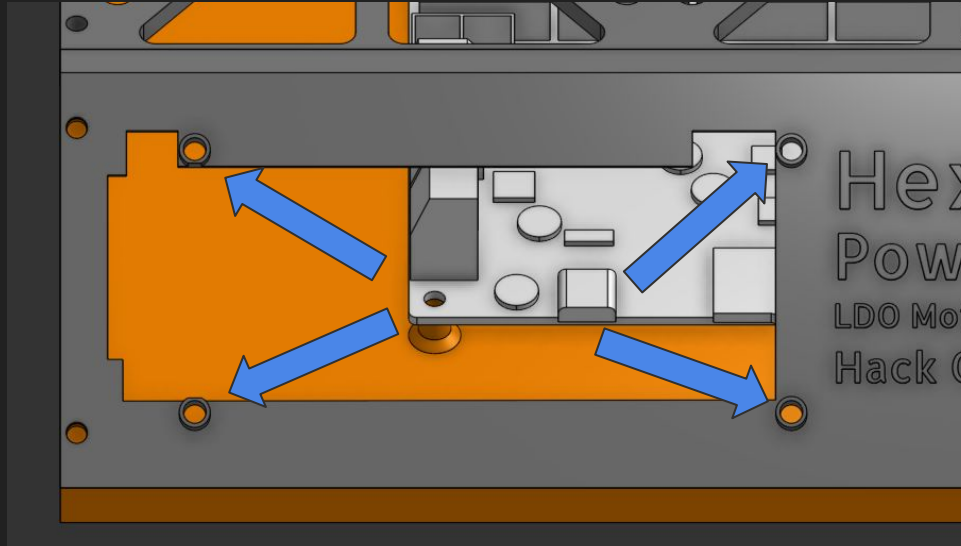


Screw the top on.



Screw the front on.

Step 4:

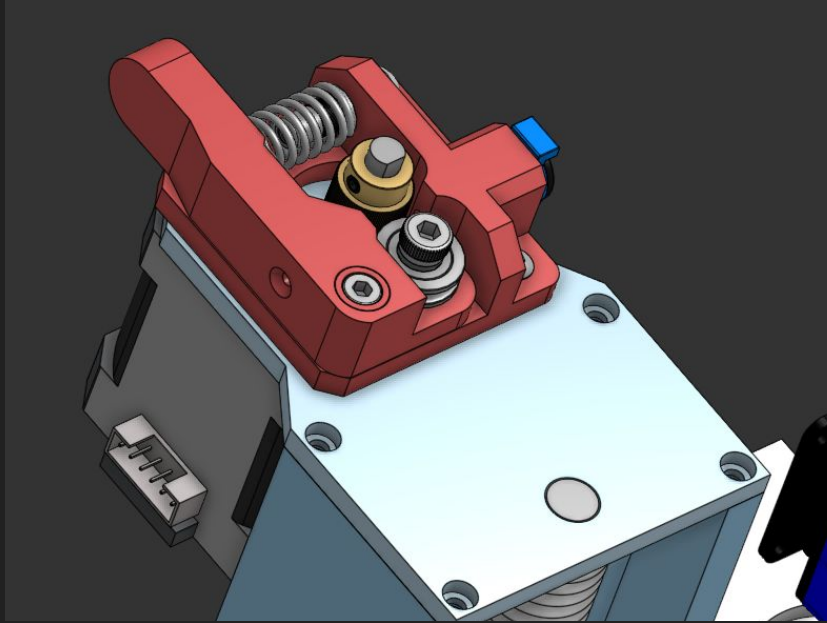


Put heated inserts where the blue arrows are



Screw the screen in (do not overtighten)

Extruder



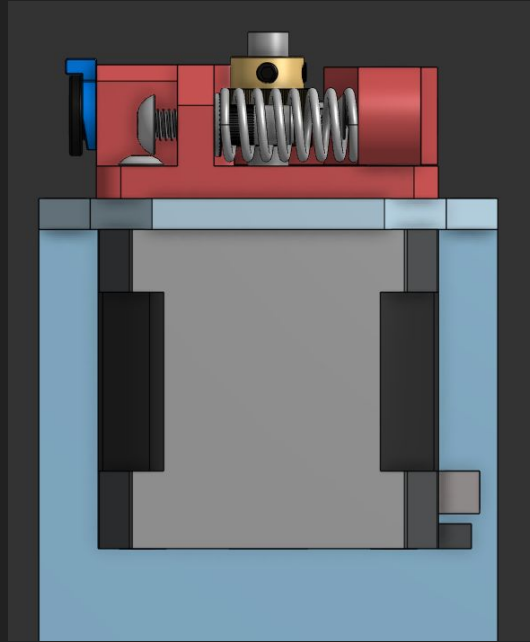
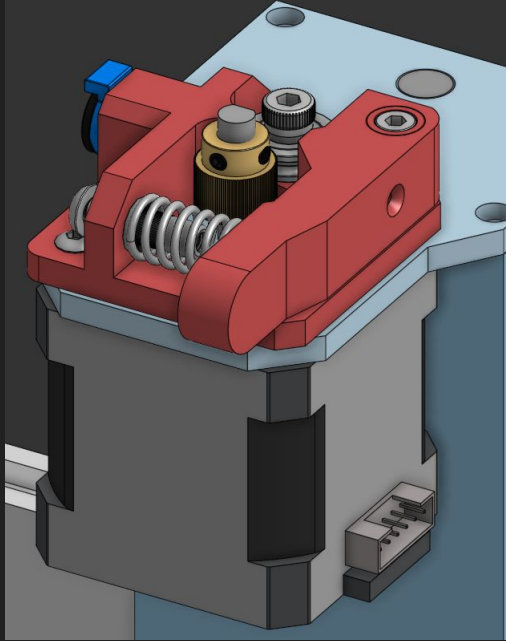
Parts Needed:

1x Nema 17

1x Bowden extruder (Ender 3) kit

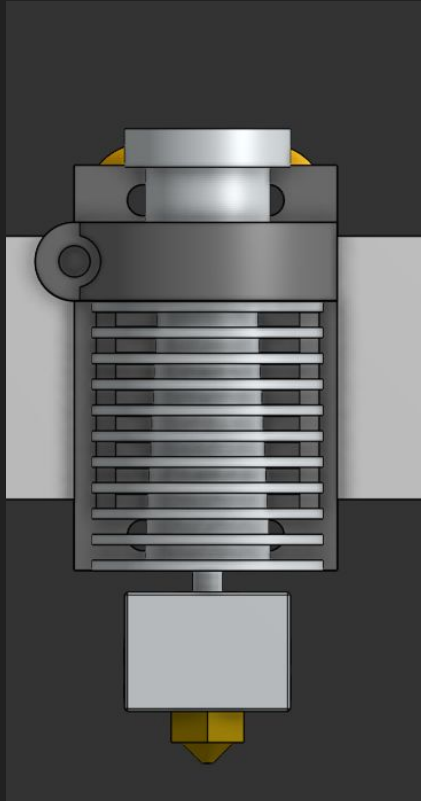
1x Bowden tubing

Step 1:



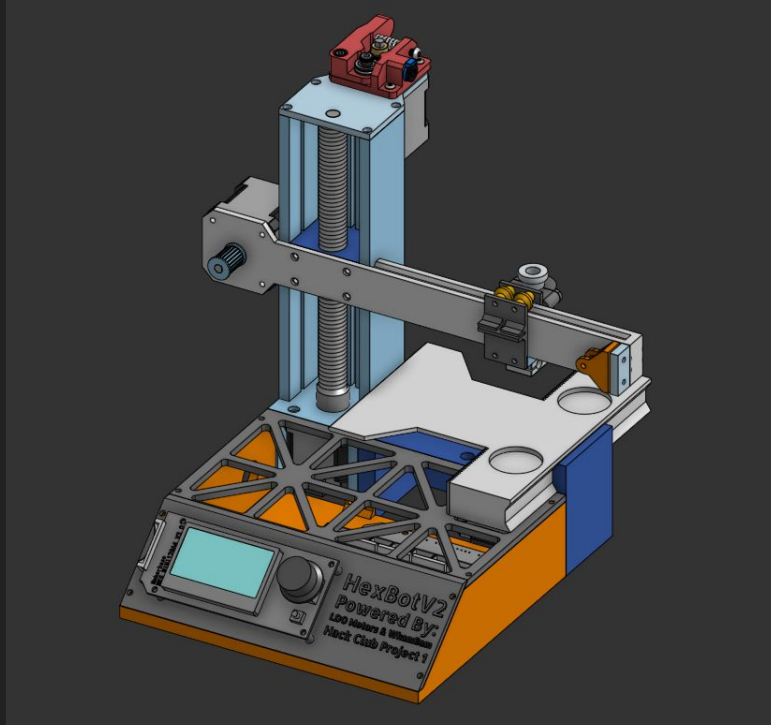
Screw the extruder onto the motor and the force should hold it good with the plastic in between.

Step 2:



Screw the screw marked by the blue arrow to clamp in the hotend.

Final assembly



Parts Needed:

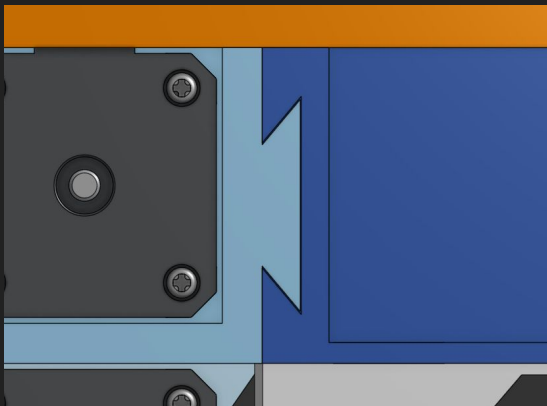
1x X-axis

1x Z-Tower

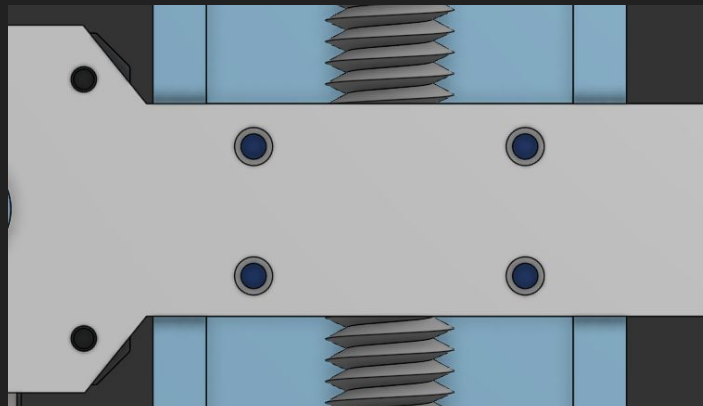
1x Y-axis

1x Base

Step 1:

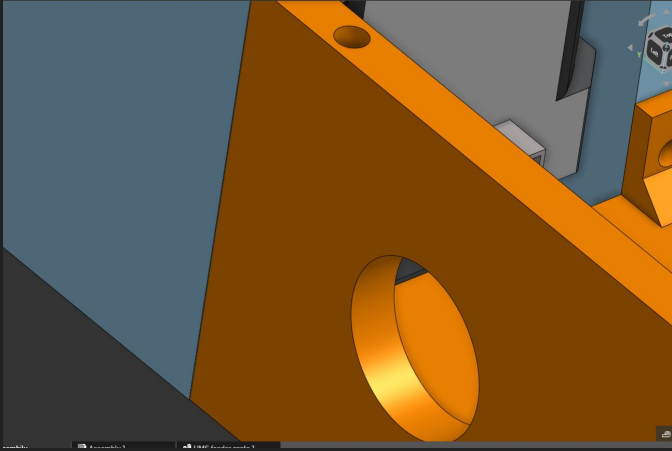


Connect the dovetail of the Z-Tower (light blue) and the Y-Axis motor mount (blue)

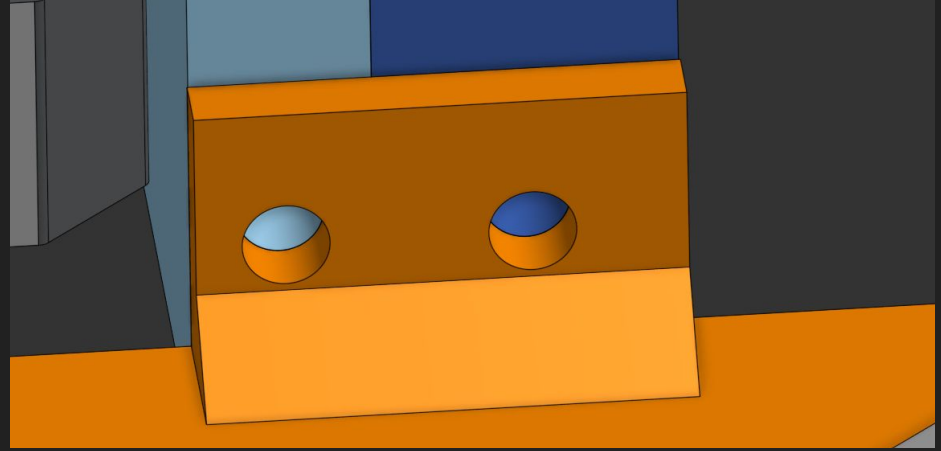


Screw these 4 holes in. If additional support is needed super glue will work or you can hot glue/ filament weld.

Step 2:



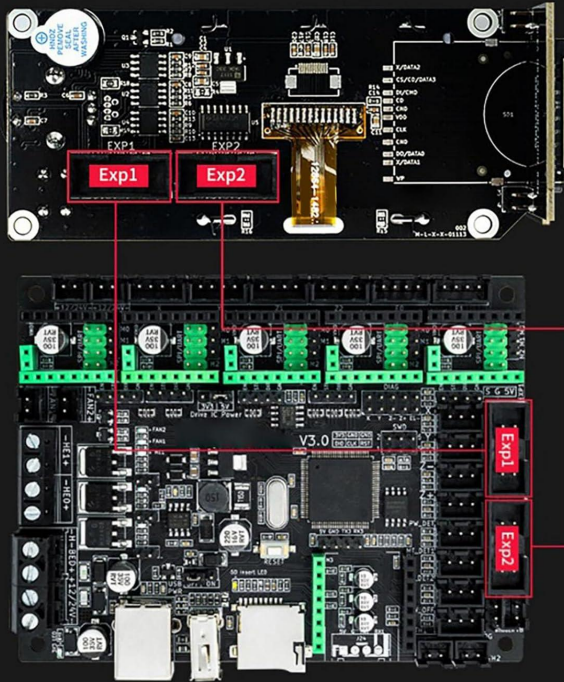
If this isn't strong enough
filament weld, superglue, or
hot glue will work



Put some screws into this

Step 3:

Connection with MKS Series Motherboard



Wire everything up.
Watch a youtube tutorial
if ur stuck. Good luck
Full BOM available on
GitHub!